MATH 205 LINEAR ALGEBRA
ASSIGNMENT 1

Q. 1

(a) Let

$$\begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{bmatrix}$$

Find A^2 and A^3.

(b) Show that $A^3 = A^2 + A - 5I$.

(c) Using (b), without doing any more multiplication, prove that

(i) $A^4 = 2A^3 - 4A^2 - 5I$,

(ii) $A^{-1} = \frac{1}{5}(I + A - A^2)$.

Q. 2

Show that the most general matrix that commutes with

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

is of the form

$$\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$$

Q. 3

(a) Let A be a square matrix, then $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric.

(b) If A is an $m \times n$ matrix, then prove that AA^T and A^TA are both symmetric (See Q.6)

Q. 3 (continued)

(c) If $A^2 = A$, A^{-1} exists, then $A = I$.

(d) If A is invertible, then prove that: $(A^{-1})^T = (A^T)^{-1}$.

Q. 4

If A is a square matrix of order 3 such that $A^T = -A$, then prove that the diagonal entries of A all equal 0.

Q. 5

Let

$$\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$$

and

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Show that if the matrix

$$\begin{bmatrix} x & y \\ z & t \end{bmatrix}$$

satisfies the equation $AX = XB$, then X is a scalar multiple of

$$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Q. 6 Important result:

If A, B are matrices such that AB is defined, then $(AB)^T = B^T A^T$.

Check this result for 2×2 matrices by taking general entries.

Note: The transpose of a product of any number of matrices is equal to the product of their transposes in the reverse order i.e.

$$(A_1 A_2 \cdots A_n)^T = A_n^T \cdots A_2^T A_1^T.$$